This Death Valley Plant Stays Cool by 'Sweating'

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California's Death Valley is one of the hottest, driest places on Earth. In summer, temperatures there often climb above 49 degrees Celsius (120 degrees Fahrenheit). Yet one shrub, the Arizona honeysweet, does more than survive there. It thrives. Now scientists have learned one of its secrets: it cools itself by, basically, sweating.

Plants release water vapor through tiny pores called **stomata** on their leaves and stems. This process is called **transpiration**. Scientists say it works a lot like sweating.

Researchers at Michigan State University wanted to know how much this cooling helps the honeysweet survive. They collected seeds from 223 wild honeysweets in and near Death Valley. In the lab, they grew more than 1,200 **seedlings** at a comfortable 30 degrees C (86 degrees F). Then they slowly turned up the heat until it reached 60 degrees C (140 degrees F) for six to eight hours a day.

Some plants survived again and again. Some of their leaves stayed about 13 degrees C (23 degrees F) cooler than the air around them. When the scientists closed the leaves' pores to shut off the cooling, the plants did not live as long.

"This cooling makes the difference between life and death," says **botanist** Seung Yon Rhee.

Not all the plants survived, though. Some lasted only a few days, while others stayed healthy for weeks. Rhee suspects the difference may come from their DNA, much like how some people have brown eyes and others have blue.

The honeysweet already amazed scientists by growing fast in extreme heat. According to MSU plant biologist Karine Prado, it does **photosynthesis** best at 47 degrees C (117 degrees F), the highest peak-performance temperature for any plant. The team shared its new findings in August. They have not yet been reviewed by other scientists.

Word Treasure

stomata -- Tiny pores on leaves and stems that let gases and water vapor in and out.

transpiration -- The way plants release water vapor through their leaves, which can cool them.

photosynthesis -- The process plants use to turn sunlight, water and carbon dioxide into food.

botanist -- A scientist who studies plants.

seedlings -- Very young plants that have just sprouted from seeds.

Background you need

Plants make their own food through photosynthesis, using sunlight, water and carbon dioxide.

They take in gases and release water vapor through tiny pores in their leaves, and that escaping water helps cool them, much as sweat cools people.

Deserts like Death Valley get very little rain and extreme heat, so plants there need special ways to survive.

Scientists study such plants to learn how living things cope with very high temperatures.

Break it down

LEAD: The Arizona honeysweet survives Death Valley heat by cooling its leaves
HOW IT WORKS: Transpiration releases water vapor through stomata, much like sweating
EXPERIMENT: Michigan State University team grew more than 1,200 seedlings and raised the heat to 60 °C
+- RESULT: Some leaves stayed about 13 °C cooler than the air
+- TEST: Closing the pores shortened the plants' lives
QUOTE: Seung Yon Rhee says the cooling makes the difference between life and death
OPEN QUESTION: Do DNA differences explain why some plants lasted longer? The findings are not yet peer reviewed

Why it matters

Learning how a plant survives some of Earth's hottest conditions helps scientists understand the limits of life and how living things cope with extreme heat.

Test what you remember

1. How hot does Death Valley often get in summer?

(A) About 20 degrees Celsius
(B) Above 49 degrees Celsius (120 degrees Fahrenheit)
(C) About 30 degrees Celsius (86 degrees Fahrenheit)
(D) Below freezing

2. What are stomata?

(A) Seeds of desert plants
(B) Tiny pores on leaves and stems
(C) Roots deep in the soil
(D) Flowers on the shrub

3. How many seedlings did the researchers grow?

(A) About 50
(B) More than 1,200
(C) 223
(D) Exactly 60

4. How much cooler than the air did some leaves stay?

(A) About 47 degrees C
(B) About 13 degrees C (23 degrees F)
(C) About 1 degree C
(D) About 60 degrees C

5. What happened when scientists closed the leaves' pores?

(A) The plants grew faster
(B) Nothing changed
(C) The plants did not live as long
(D) The plants changed color

6. What does Seung Yon Rhee suspect explains why some plants lasted longer?

(A) How much sunlight the lab had
(B) Differences in their DNA
(C) The age of the seeds' collectors
(D) The color of their pots

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: C 6: B

Questions to think about

1. Seung Yon Rhee, botanist: Says the leaf cooling makes the difference between life and death and suspects DNA explains survival differences.

2. Karine Prado, MSU plant biologist: Says the honeysweet has the highest peak-performance temperature for photosynthesis of any plant.

MY THOUGHTS (at least 20 words):